

Ethernet Port

Raspberry Pi Model A+ came with no Ethernet slot. Being the first model, they had a lot to learn and teach. Since then, all the models of Raspberry Pi except the Zero series and 3 Model A+ came with a 100 Base-T Ethernet support for all the users.

Bluetooth and Wi-Fi

Bluetooth and Wi-Fi were added to the Raspberry Pi line-up starting with the Raspberry Pi 3. It came with Bluetooth 4.1 LE and support for 802.11n Wi-Fi routers. The same configuration was present in Raspberry Pi Zero W and WH as well. The Raspberry Pi 3 A+ and B+ saw BLE support added for the Bluetooth connectivity and full-fledged 2.4 GHz and 5 GHz 802.11 b/g/n/ac for Wi-Fi. With Raspberry Pi 4, the Bluetooth has up to the latest 5.0, and the Wi-Fi support remains the same.

Camera Modules

The Raspberry Pi has support for the camera module, which can click still pictures or video recordings. The camera module connects to the CSI-2 camera port via a short ribbon cable. The module also connects to the ISP present in the SoC, which processes the incoming data and outputs the final format as required.

Initially, only one camera module was supported, the Omnivision OV5647 sensor. Users can now also connect the Sony IMX219 sensor as well, which is a sensor commonly-used for smartphone photography and is quite capable. Users can use the module to shoot JPEGs, PNGs, GIFs, and BMP along with several other formats that are supported. For video H.264, baseline, main, and high-profile formats are compatible with the Pi.

Arduino

Arduino is an open-source platform that helps in creating various electronic projects. Unlike Raspberry Pi, Arduino is used in varied industries and for multiple purposes. The biggest reason why Arduino is gaining traction is that it doesn't require a programmer to run codes. A USB cable is all you need to load new codes on to your device.

Unlike Raspberry Pi, Arduino is mostly used to build cheap scientific instruments or for beginners in the field of Robotics and programming. It is an easy-to-operate tool for you to try your weird ideas and to make them a success.